

Lecture 08

Synthetic Streamflow Generation

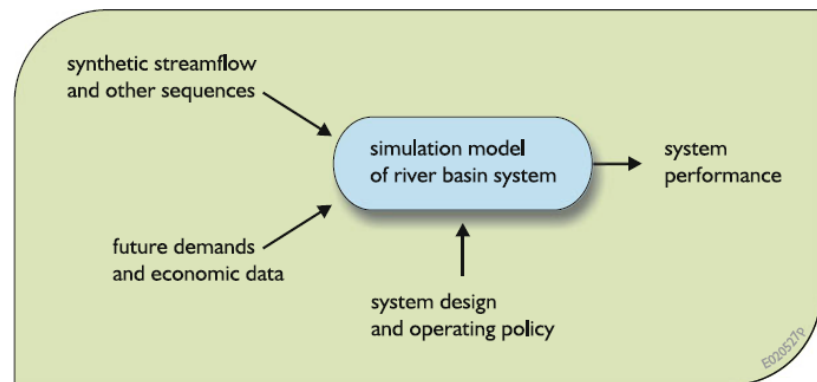
6.8 Synthetic Streamflow Generation (p.257)

❖ Why do we Generate Synthetic Streamflow?

- ◆ Little or no data for your system
- ◆ Evaluate the effect of hydrologic non-stationarity on your system

❖ Types of Synthetic streamflow generators

- ◆ Indirect approach: generate precipitation and temperature series and input them to a hydrologic model of a basin (GCM)
- ◆ Direct approach: uses statistical techniques to generate streamflow time series directly



6.8 Synthetic Streamflow Generation (p.257)

❖ Types of Synthetic streamflow generators

- ◆ **Direct approach**: uses statistical techniques to generate streamflow time series directly

❖ Types of direct approaches

◆ Parametric methods

- Relies on hypothesized statistical model of streamflow whose parameters are selected to achieve a desired result
- Time series model! (Autoregressive Moving Average- ARMA)

◆ Non-parametric methods

- Re-sample flows from a record & rely heavily on the data
- Examples include Block bootstrapping, k-nearest neighbor, etc.

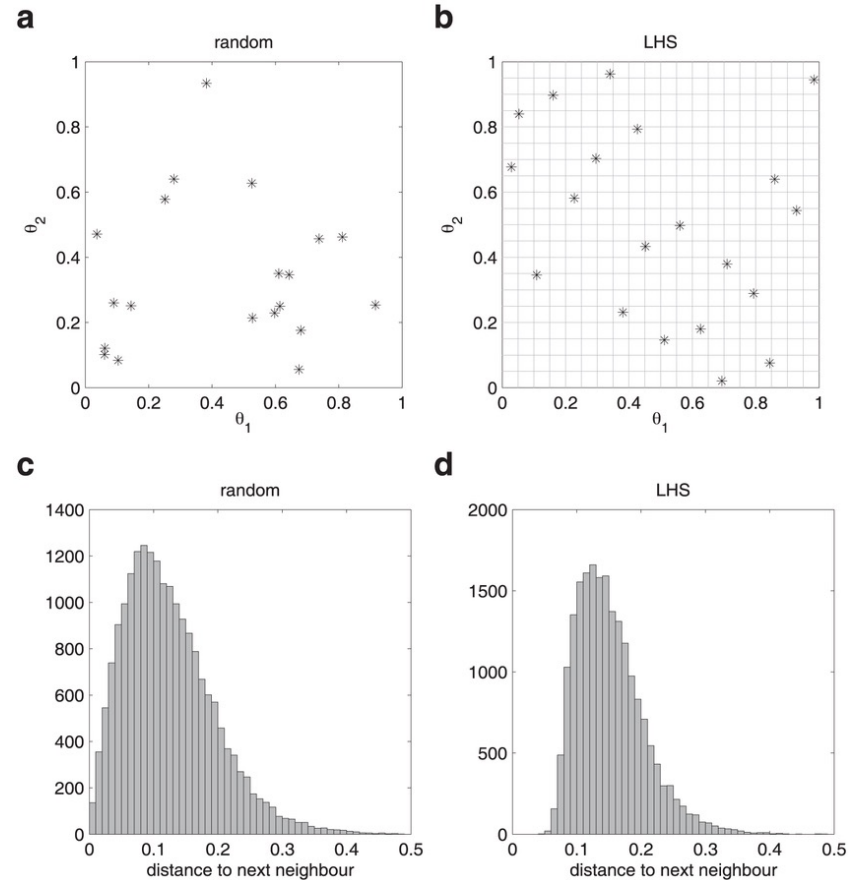
Common Sampling Methods

❖ Monte Carlo Sampling (MC)

- ◆ 무작위 (전 시행과 독립적으로)로 변수를 선택
- ◆ 실제 분포와 유사한 결과를 위해서는 LHS보다 더 많은 반복을 거쳐야 함 (예시-Excel)

❖ Latin Hypercube Sampling (LHS)

- ◆ 각 변수의 확률을 겹치지 않는 Segment로 구분
- ◆ 전체 범위의 분포가 보다 균일하고 일관되게 샘플링 됨



Application in the Real-world

- ❖ 만약 예측하고자 하는 변수의 평균, 분산, Lag-1 Correlation 등의 기초 통계량이 시간에 따라 변하지 않고,
- ❖ 예측하고자 하는 변수가 실제로 가정한 분포를 완벽하게 따른다면 위의 Sampling Methods를 실제 분석에 사용해도 무방하나
- ❖ 그러나 현실에서 이러한 가정을 만족시키는 변수를 찾기는 어렵.
- ❖ Streamflow Generation Models (from p. 260)
 - ◆ A Simple Autoregressive Model
 - ◆ Multivariate Models (General multisite streamflow model)

Types of Streamflow Generation Models

❖ Streamflow Generation Models (from p. 260)

◆ A Simple Autoregressive Model

- Based on the Autoregressive Markov model ($Q_{y+1} \sim Q_y$)

$$Q_{y+1} = \mu + \rho(Q_y - \mu) + V_y \sigma \sqrt{1 - \rho^2}$$

◆ Multivariate Models (General multisite streamflow model)

- Fitting time series models to each side separately
then correlating the two models to reproduce the cross-correlation
between the series

◆ Multi-season, Multisite Models

- Disaggregation Models: Annual -> Seasonal
- Aggregation Models: Seasonal -> Annual

SAMS2007

❖ Stochastic Analysis Modeling and Simulation (SAMS 2007)

<http://www.sams.colostate.edu/>

Stochastic Analysis Modeling and Simulation
SAMS 2007
O.G.B. Sveinsson, J.D. Salas, W.L. Lane and D.K. Frevert

Colorado State University

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WELCOME!

SAMS is a computer software package that deals with the stochastic analysis, modeling, and simulation of hydrologic time series such as annual and monthly streamflows. It has been developed as a cooperative effort between **Colorado State University** and the **US Bureau of Reclamation** in Denver, Colorado.

SAMS has been written in C++ and Fortran and runs under modern windows operating systems such as WINDOWS 2000, WINDOWS NT, and WINDOWS XP. The package consists of many menu windows that enables the user to choose among various options that are currently available. **SAMS** has the capability of analyzing, modeling and generating single site and multisite annual and seasonal data and the results of the analysis are presented in graphical or tabular forms or are written on output files.

The current version of SAMS, denoted as **SAMS-2007**, may be applicable for annual and seasonal data such as quarterly and monthly data. SAMS allows for *three basic options*: 1) Statistical Analysis of Data, 2) Fitting a Stochastic Model (includes parameter estimation and testing), and 3) Generating Synthetic series. Further details about SAMS may be obtained from Overview and SAMS Description as shown below.

- Overview
- SAMS Description
- Documentation (Manual)
- SAMS Exec Download
- SAMS Applications

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SAMS2007

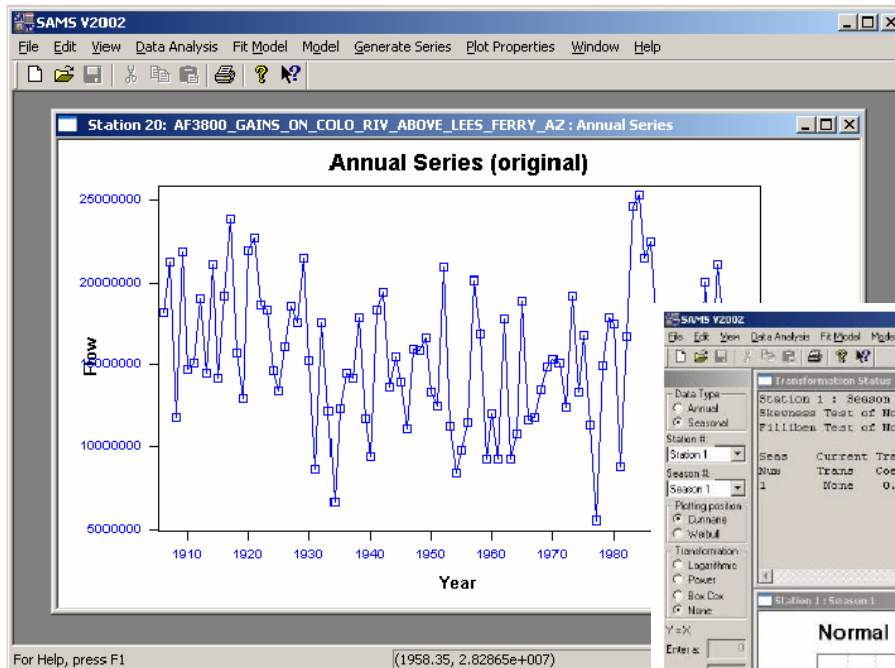


Figure 2.5 Time series of the annual flows of the Colorado

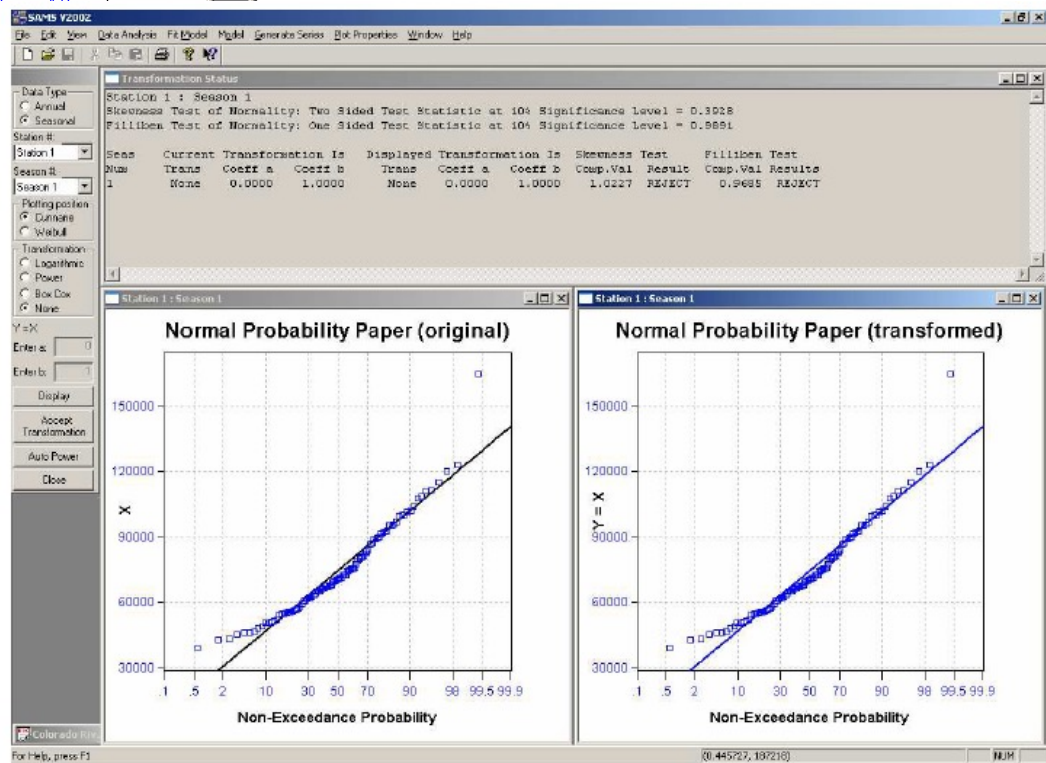


Figure 2.6 Plot of the data on normal probability paper and test of normality

SAMS2007

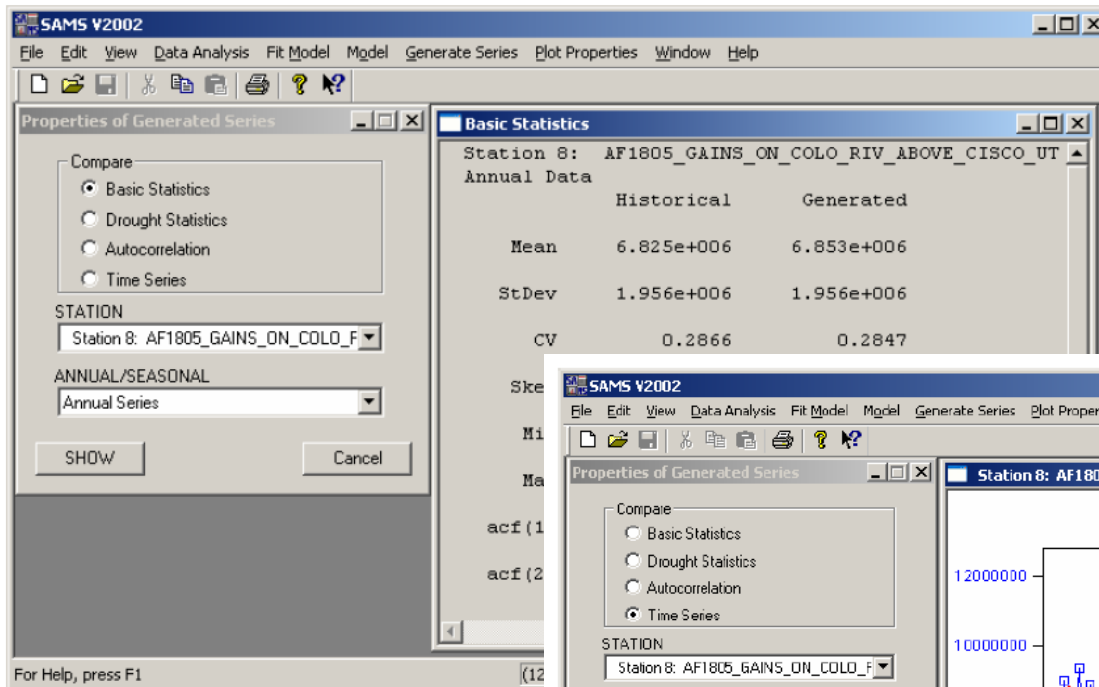


Figure 2.27: Comparison of the basic statistic

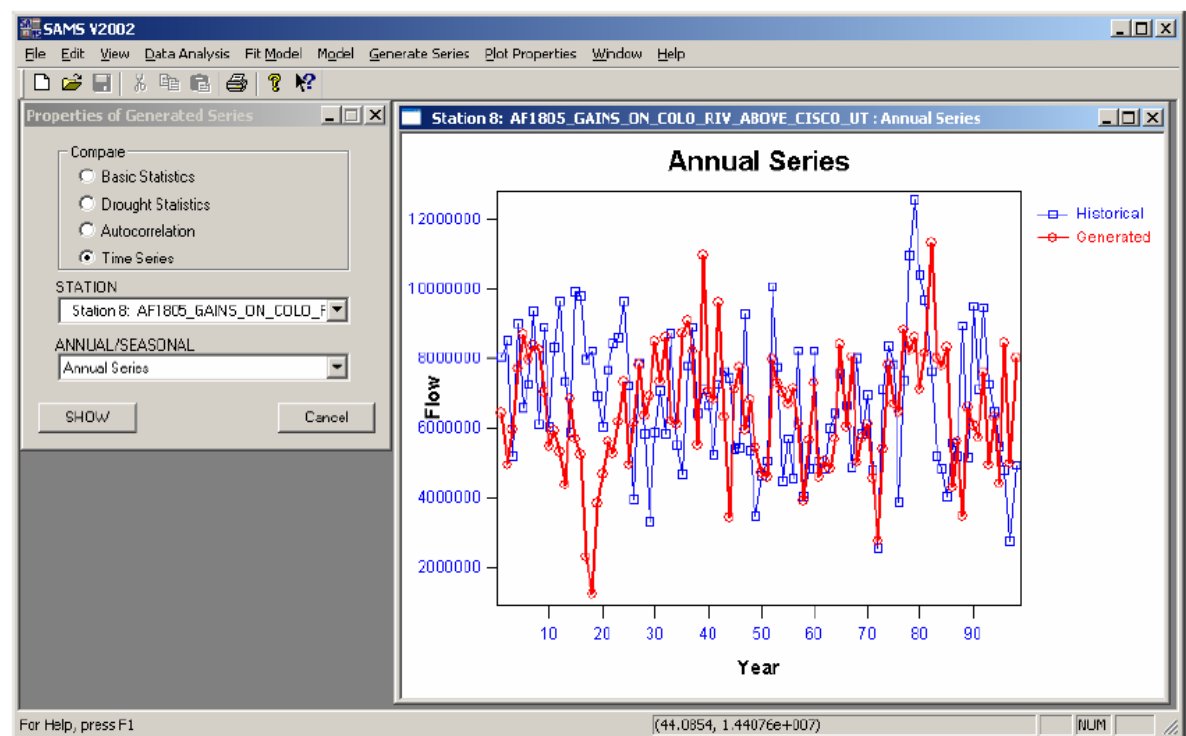


Figure 2.28: Comparison of the time series.

SAMS2007

4 MATHEMATICAL MODELS

The following univariate and multivariate models are available in SAMS for modeling of annual and seasonal data.

1. For Annual Modeling:

- Univariate ARMA(p,q) model.
- Univariate GAR(1) model.
- SM (shifting mean) model.
- Multivariate AR(p) model (MAR).
- Contemporaneous ARMA(p,q) model (CARMA(p,q)).
- Mixture of contemporaneous shifting mean and ARMA(p,q) models (CSM – CARMA(p,q)).

2. For Seasonal Modeling:

- Univariate PARMA(p,q) model.
- Multivariate PAR(p) model (MPAR).

3. Disaggregation Models

- Spatial Valencia and Schaake.
- Spatial Mejia and Rousselle.
- Temporal Lane.
- Temporal Grygier and Stedinger.

End of Lecture 08

❖ May-31 (Today's Lecture)

◆ Lec 08: Synthetic Streamflow Generation

- A Simple Autoregressive Model
- Multivariate Models (General multisite streamflow model)
- Multi-season, Multisite Models
(Disaggregation Models & Aggregation Models)
- Introduction to **SAMS2007**
(Stochastic Analysis Modeling and Simulation)

◆ Lec 09: Water Resources Systems Engineering Frontiers -Introduction and Examples Part 1

~~◆ Lec 10: Water Resources Systems Engineering Frontiers -Introduction and Examples Part 2 (*Invited Talk*)~~

<https://www.youtube.com/watch?v=tnRCNHsk69I>

기타 공지

❖ Term Project B 팀 변경

- ◆ 팀 C: 이재황 김현주 박성열 남현욱
- ◆ 팀 D: 이재호 김연주 이주형

❖ 강의평가 관련

- ◆ 2022.06.03. (금) 5PM 이전까지 팀원 전체가 강의평가를 완료한다면 Term Project B 점수에 팀별 점수 +1점
- ◆ 강의평가는 다음 장 내용을 이메일이나 카카오톡(개인메세지)으로 전달
- ◆ 학점 확인을 위한 강의평가는 비슷한 내용으로 향후에 진행해도 무방

2022 Spring WRSE 강의평가 문항

- ❖ 2022년 1학기 수자원시스템공학 수업과 관련해서 아래 질문들에 대해 최대한 자세하게 작성 부탁드립니다. 여러분의 자세한 피드백이 수업 개선에 큰 도움이 됩니다 (1-3 문항별 공백 포함 200자 이상).
- ◆ 이번 수업에서 가장 좋았던/괜찮았던 내용은 무엇이었습니까?
 - ◆ 수업 내용 중, 가장 어려웠던/힘들었던 부분은 어떤 것이었습니까?
 - ◆ 수업 관련한 건의사항이 있으면 말씀해 주세요 (강의 시간, 강의 내용, 퀴즈나 숙제 관련, 시험 관련, 텀프로젝트, 평가 방법 등).
 - ◆ (있을 시, 분량 상관없이 답변) 수업 내용과 관련하여 추가적으로 수업에서 다루어졌으면 하는 내용이 있으신가요?
 - ◆ (있을 시, 분량 상관없이 답변) 강의자에게 따로 전달할 메시지나 건의사항이 있으면 (수업 외 내용도 괜찮습니다) 따로 전달 부탁드립니다.